

SAHIL YADAV

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SUMMARY

Applied mathematician and ML engineer specializing in **large-scale optimization and decision systems under uncertainty** – MILP, stochastic & constrained optimization, forecasting, and deep learning. Track record of shipping models with direct, multi-crore P&L impact on live operational systems. M.Sc. Mathematics + B.E., BITS Pilani; published research in deep learning and GANs.

EDUCATION

Birla Institute of Technology and Science, Pilani

2018 - 2023

M.Sc. (Hons.) Mathematics + B.E. (Hons.) Electrical & Electronics Engineering

WORK EXPERIENCE

Zepto, Bangalore

Jul 2023 – Present

Data Scientist I → II → Senior Data Scientist

- Designed an end-to-end **MILP optimization engine** for vehicle scheduling & routing – full dock-to-store-to-return timetables with second-trip batching under capacity, manpower, timing & no-overlap constraints; **5 Cr cost reduction** nationwide.
- Built a **combinatorial pick-path optimization system** via co-occurrence-based placement and affinity-driven sequencing, cutting **average travel distance ~30%** (rolling out).
- Engineered a large-scale **demand forecasting engine** (gradient-boosted, custom loss) over high-dimensional SKU data, delivering a **5% MAPE / 10% Bias reduction** and a **22 Cr reduction in fortnightly procurement**.
- Led cross-functional development of an **Auto-Replenishment System** a high-performance warehouse-to-store transfers generator with low-latency planning and reduced manual overhead, **yielding a 10% availability uplift**.
- Guided an **assortment recommendation model** fusing search-to-conversion signals, relevance scoring & competitor data to surface emerging demand.

Zepto, Mumbai - Data Science Intern

Summer 2022

- Formulated a **PID control-based dynamic pricing model** balancing real-time demand against supply stress while minimizing order loss.

Pixis AI, Bangalore - Data Science Intern

Fall 2021

- Built a **relevance ranking model** over video metadata using embedding similarity and NLP to improve ad conversion.

Google Summer of Code (GSoC) 2020 - OpenAstronomy

Summer 2020

- Trained **RNN-based models** for gamma-ray event reconstruction & classification and integrated ROOT data pipelines for the MAGIC & VERITAS telescopes.

TECHNICAL SKILLS

Optimization: MILP, constrained & stochastic optimization, operations research, control systems

ML / Math: forecasting, gradient boosting, deep learning, GANs, RNNs, probability & statistics

Engineering: Python, PySpark, SQL, large-scale data pipelines

PUBLICATIONS

- S. Yadav, A. Mehra, H. Rohmetra, R. Ratnakumar, P. Narang. **DerainGAN: Single Image Deraining Using Wasserstein GAN**. *Multimedia Tools and Applications*, Springer
- T. Miener et al. (incl. S. Yadav). **IACT event analysis with the MAGIC telescopes using deep convolutional neural networks with CTLearn**. *ADASS XXXI*